

CHEMISTRY

Sections A – D

Solutions · Chemical Kinetics · Electrochemistry · Haloalkanes and Haloarenes · Alcohols, Phenols and Ethers · Biomolecules

Time allowed: not stated on the original

Maximum Marks: 62

General Instructions

1. Answer **all** questions.
2. The intended marks for each question, or part of a question, are given in brackets [].
3. Where a question is followed by **OR**, attempt **either** the question **or** its alternative, not both.
4. All working, including rough work, should be done on the same sheet as the rest of the answer.
5. Balanced equations and structural formulae should be given wherever required.

SECTION A — 15 MARKS

Question 1 Answer **all** fifteen sub-parts.

[15 × 1 = 15]

(i) Conversion of chlorobenzene into phenol involves:

- (a) modified S_N1 mechanism
- (b) modified S_N2 mechanism
- (c) both (a) and (b)
- (d) elimination–addition mechanism

(ii) Phenol on treatment with excess of conc. HNO_3 gives:

- | | |
|--|---------------------------|
| (a) <i>o</i> -nitrophenol | (b) <i>p</i> -nitrophenol |
| (c) <i>o</i> - and <i>p</i> -nitrophenol | (d) 2,4,6-trinitrophenol |

(iii) CH_3CH_2OH can be converted into CH_3CHO by:

- | | |
|--------------------------------------|---------------------------------|
| (a) catalytic hydrogenation | (b) treatment with $LiAlH_4$ |
| (c) treatment with PCC in CH_2Cl_2 | (d) treatment with $KMnO_4/H^+$ |

(iv) Which mixture shows positive deviation from Raoult's law?

- | | |
|-------------------------|--------------------------|
| (a) Acetone + Aniline | (b) Acetone + Ethanol |
| (c) Water + Nitric acid | (d) Chloroform + Benzene |

(v) Charge required to liberate 1 mol of $MnO_4^- \rightarrow Mn^{2+}$ is:

- | | |
|---------|---------|
| (a) 1 F | (b) 3 F |
| (c) 5 F | (d) 4 F |

(vi) Henry's law constant is larger when gas solubility is:

- | | |
|-----------------|------------------------|
| (a) Higher | (b) Lower |
| (c) Independent | (d) Pressure-dependent |

(vii) Deficiency of which vitamin causes rickets?

- | | |
|---------------|---------------|
| (a) Vitamin C | (b) Vitamin B |
| (c) Vitamin A | (d) Vitamin D |

(viii) The values of van't Hoff factor for KCl , K_2SO_4 and $C_6H_{12}O_6$ are:

- | | |
|----------------|----------------|
| (a) 2, 2 and 2 | (b) 2, 3 and 2 |
| (c) 3, 2 and 1 | (d) 2, 3 and 1 |

(ix) The reaction of toluene with Cl_2 in presence of FeCl_3 gives 'X', while the reaction of toluene with Cl_2 in presence of light gives 'Y'. Thus 'X' and 'Y' are:

- (a) X = benzyl chloride, Y = *o*- and *p*-chlorotoluene
- (b) X = *m*-chlorotoluene, Y = *p*-chlorotoluene
- (c) X = *o*- and *p*-chlorotoluene, Y = trichloromethylbenzene
- (d) X = benzyl chloride, Y = *m*-chlorotoluene

Sub-parts (x) to (xiv) are assertion–reason questions. For each, select the correct option from those given below.

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true but R is false.
- (d) A is false but R is true.

Note. The original paper omits this option list entirely. It is supplied here, in the standard form, so that the six assertion–reason marks can actually be attempted.

(x) **Assertion (A):** *p*-nitrophenol is more acidic than phenol.

Reason (R): The $-\text{NO}_2$ group helps in the stabilisation of the phenoxide ion by dispersal of negative charge due to resonance.

(xi) **Assertion (A):** Henry's law constant increases with temperature.

Reason (R): Solubility of a gas in a liquid decreases with increase in temperature.

OR

Assertion (A): Isotonic solutions have the same osmotic pressure.

Reason (R): 0.1 M NaCl and 0.1 M glucose will be isotonic.

(xii) **Assertion (A):** Uracil is found in RNA.

Reason (R): RNA does not undergo replication.

(xiii) **Assertion (A):** The rate and the rate constant of a zero order reaction have the same units.

Reason (R): The half-life of a zero-order reaction is independent of the initial concentration of the reactant.

(xiv) **Assertion (A):** Reactions happen faster at higher temperatures.

Reason (R): As temperature increases, there is also an increase in the number of collisions with the required activation energy for a reaction to occur.

(xv) For the reaction $\text{A} \rightarrow \text{B}$, the rate of reaction is quadrupled when the concentration of A is doubled. If the rate of the reaction is $r = [\text{A}]^n$, the value of n is:

- (a) 1
- (b) 0
- (c) 3
- (d) 2

SECTION B — 14 MARKS

1. Write down the mechanism for the acid catalysed **dehydration** of ethanol to ethene.

[2]

Note. The original reads "the acid catalysed of ethanol to ethene"; the missing word is **dehydration**.

OR

Carry out the following conversions in not more than 2 steps:

[2]

- a) But-1-ene to But-2-ene
- b) Benzene to Biphenyl

2. Calculate the mass of ascorbic acid (Vitamin C, $\text{C}_6\text{H}_8\text{O}_6$) to be dissolved in 75 g of acetic acid to lower its freezing point by 1.5°C . [$K_f = 3.9 \text{ K kg mol}^{-1}$]

3. A first-order reaction is 50% completed in 40 minutes at 300 K and in 20 minutes at 320 K. Calculate the activation energy of the reaction. [2]

[Given: $\log 2 = 0.3010$, $\log 4 = 0.6021$, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$]

OR

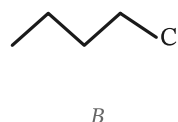
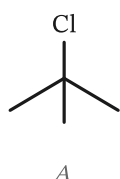
Define order of reaction. How does the order of a reaction differ from molecularity for a complex reaction? [2]

4. The rate constant for a first order reaction is 60 s^{-1} . How much time will it take to reduce the initial concentration of the reactant to 1/10th of its initial value? [2]

OR

After 24 hrs only 0.125 g out of the initial quantity of 1 g of a radioactive isotope remains behind. What is its half-life period? [2]

5. In the following pair of halogen compounds, which compound undergoes the faster $\text{S}_{\text{N}}1$ reaction? [2]



Note. Redrawn as skeletal formulae from the hand-drawn structures on the original.

6. Write down the balanced chemical equation for the following name reactions: [2]

- a) Kolbe's reaction
- b) Williamson's ether synthesis

7. Name the reagent used in the following reactions: [2]

- a) Benzyl alcohol to benzoic acid
- b) Oxidation of a primary alcohol to a carboxylic acid

SECTION C — 18 MARKS

8. [3]

- a) Differentiate between globular and fibrous protein.
- b) Give one example each of an essential and a non-essential amino acid.

OR

Which sugar is called invert sugar? Why is it called so?

Define peptide linkage.

9. [3]

- a) The slope of the line in the graph of $\log k$ against $1/T$ for a reaction is -5841 K . Calculate the energy of activation.
- b) Define temperature coefficient.

OR

The rate of a reaction depends upon the temperature and is quantitatively expressed as $k = Ae^{-E_a/RT}$

- a) If a graph is plotted between $\log k$ and $1/T$, write the expression for the slope.
- b) Under different conditions E_{a1} and E_{a2} are the activation energies of two reactions. If $E_{a1} = 40 \text{ J/mol}$ and $E_{a2} = 80 \text{ J/mol}$, which of the two has the larger value of the rate constant?

10. Give reasons for the following: (any three)

[3]

- (i) Alkyl halides, though polar, are immiscible with water.
- (ii) A racemic mixture is optically inactive.
- (iii) Grignard reagents should be prepared under anhydrous conditions.
- (iv) Ethyl iodide undergoes S_N2 reaction faster than ethyl bromide.
- (v) *Ortho*-nitrophenol has a lower boiling point than *para*-nitrophenol.

11. 2 g of benzoic acid (C_6H_5COOH) dissolved in 25 g of benzene shows a depression in freezing point equal to 1.62 [3]
K. The molal depression constant for benzene is 4.9 K kg mol^{-1} . What is the percentage association of the acid if it forms a dimer in solution?

12. [3]

- (a) **Explain why** $(CH_3)_3C-O-CH_3$ on reaction with HI gives CH_3OH and $(CH_3)_3C-I$ as the main products, and not $(CH_3)_3C-OH$ and CH_3I .
- (b) Give a chemical test to distinguish between the following compounds:
 - i) Ethanol and *tert*-butyl alcohol
 - ii) Ethanol and methanol

Note. In the original, part (a) prints the statement with no instruction. The instruction **Explain why** has been supplied.

13. [3]

- (a) Give chemical tests to distinguish between the following pairs of compounds:
 - (i) Pentan-2-ol and Pentan-3-ol
 - (ii) Methanol and Phenol
- (b) Why are aromatic ethers *o,p*-directing?

OR

Convert:

- i) Aniline to phenol
- ii) Ethyl magnesium chloride to propanol
- iii) Phenol to picric acid

SECTION D — 15 MARKS

14. [5]

- a) An organic compound 'A' having molecular formula C_3H_6 , on treatment with aqueous H_2SO_4 , gives 'B', which on treatment with conc. $HCl/ZnCl_2$ gives 'C'. Compound 'C' on treatment with ethanolic KOH gives back the compound 'A'. Identify A, B and C.
- b) How can propan-2-one be converted into *tert*-butyl alcohol?

15. [5]

- (a) For the reaction $2A + B \rightarrow C + D$, the following results have been obtained during kinetic studies:

Exp.	[A] / mol L^{-1}	[B] / mol L^{-1}	Initial rate / $\text{mol L}^{-1} \text{s}^{-1}$
1	0.1	0.1	6.0×10^{-3}
2	0.3	0.2	7.2×10^{-2}
3	0.3	0.4	2.88×10^{-1}
4	0.4	0.1	2.40×10^{-2}

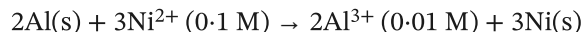
Determine the rate law and the rate constant for the reaction.

Note. The original prints experiment 2 as 7×10^{-2} , which is inconsistent with the other three rows; the intended value 7.2×10^{-2} is printed here.

(b) The conversion of the molecule X to Y follows second order kinetics. If the concentration of X is increased to three times, how will it affect the rate of formation of Y?

16. [5]

a) Represent the cell in which the following reaction takes place:



Calculate its emf if $E^\circ_{\text{cell}} = 1.41 \text{ V}$.

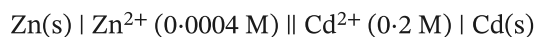
b) (i) State Kohlrausch's law of independent migration of ions.

(ii) State two advantages of the $\text{H}_2\text{-O}_2$ fuel cell over an ordinary cell.

OR

(a) The emf of a galvanic cell composed of two hydrogen electrodes is 0.16 volt at 25°C . Calculate the pH of the anode solution if the cathode is in a solution with $\text{pH} = 1$.

(b) Calculate the cell emf and ΔG° for the cell reaction at 25°C for the cell:



$[E^\circ \text{ at } 25^\circ\text{C}: \text{Zn}^{2+}/\text{Zn} = -0.763 \text{ V}; \text{Cd}^{2+}/\text{Cd} = -0.403 \text{ V}; F = 96500 \text{ C mol}^{-1}; R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}]$

END OF PAPER

HOW THIS DIFFERS FROM THE SCANNED ORIGINAL

1. The **assertion–reason option list**, which the original never prints, is supplied before sub-part (x).
2. **Q15, experiment 2** reads 7.2×10^{-2} , not the 7×10^{-2} printed on the original: with 7.0 no single rate law fits all four rows.
3. **Q12(a)** has been given its missing instruction, “Explain why”.
4. **Section B Q1** has been given its missing word, “dehydration”.
5. The **two structures in Q5** are redrawn as skeletal formulae, and scanner typography is corrected ($\text{LiAlH}_4 \rightarrow \text{LiAlH}_4$, $\text{KMO}_4 \rightarrow \text{KMnO}_4$, $\text{Cl}_2 \rightarrow \text{Cl}_2$, $\text{S}_\text{N}1 \rightarrow \text{S}_\text{N}1$, “emi” $\rightarrow$ emf).
6. Question **numbering is unchanged**: Section B restarts at 1 and Section C continues from 8, exactly as printed. The original states no time allowed and gives no general instructions; the instructions above are supplied.